

Vol 6 — Thermodynamics & Statistical Mechanics from D_IV⁵

BST Physics Curriculum | Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate)

Status: SCAFFOLD v0.1 (Saturday 2026-05-23 Wave 2) — chapter content TO BUILD Lead: Lyra (primary, thermo-as-universal-substrate-process per Casey insight) + Elie (computational verification)

Reader target

Undergraduate + graduate physics. Reads Vol 0 (Substrate Foundation) as prerequisite. Restructured volume: thermodynamics LEAD (not stat mech afterthought) per Casey’s insight — entropy + temperature + free energy are universal substrate processes; statistical mechanics derives from substrate-tick $GF(128)^k$ discretization. Heat kernel $\times$ partition function as load-bearing chapter.

Prerequisites

- Vol 0 Substrate Foundation
- Vol 1 Ch 5 Casimir Algebra + Ch 6 Operator zoo + Ch 7 Dynamics
- Vol 4 Ch 4 Λ from Substrate (Casimir vacuum cross-link)
- Standard thermodynamics + statistical mechanics undergraduate background

BST integer focus

Thermodynamics + Statistical Mechanics in BST is substrate-process language for the substrate’s per-tick computation: - Entropy = Shannon entropy of substrate-tick $GF(128)^k$ state distribution per Reed-Solomon discretization (Casey Wednesday “entropy = force = counting at depth 0”) - Temperature = substrate-tick computational rate $\times$ Casimir-eigenvalue spread (substrate-natural temperature) - Free energy = substrate’s per-tick committed-vs-residual mode balance (T2106 + Vol 4 Ch 2 eigentone framework) - Partition function Z = heat kernel evaluation on Bergman $H^2(D_{IV}^5)$ at substrate-temperature scale (load-bearing cross-link Vol 6 Ch 5)

Chapter outline (10-12 chapters, TO BUILD)

Ch	Title	Reader takeaway	Cross-link
1	Thermodynamics from Substrate	Entropy + temperature + free energy as substrate processes	Casey “entropy = force = counting”

Ch	Title	Reader takeaway	Cross-link
2	Heat + Work + First Law	Substrate Zone-1 absorption + Zone-3 emission framing	T2417 4-Zone Commitment Cycle
3	Entropy + Second Law	Shannon entropy of substrate-tick $GF(128)^k$ state distribution	Vol 1 Ch 10 substrate-tick + Vol 7 Information Theory
4	Free Energies + Maxwell Relations	Substrate committed vs residual mode balance	T2106 + Vol 4 Ch 2
5	Partition Function + Heat Kernel (LOAD-BEARING)	Z = heat kernel evaluation on Bergman $H^2(D_{IV}^5)$; cross-link to T1485 + Λ derivation	Vol 4 Ch 4 + T1485 + heat-kernel chain T_k speaking pairs
6	Classical Statistical Mechanics	Boltzmann + Maxwell-Boltzmann distributions as substrate-equilibrium emergent	Standard classical stat mech recovery
7	Quantum Statistical Mechanics	Bose-Einstein + Fermi-Dirac distributions from substrate $Pin(2)$ Z_2 grading	Paper #133 spin-statistics + Vol 5 Ch 9
8	Phase Transitions	Substrate phase transitions (T_c BBN scale, T2405 Koons tick)	Vol 4 Ch 8 BBN + Vol 4 Ch 7 inflation T_c
9	Critical Phenomena + Renormalization Group	Substrate-tick cyclotomic chain = 7-step RG flow (K59 RATIFIED)	Vol 1 Ch 10
10	Casimir Effect + Vacuum Thermodynamics	Laboratory Casimir + cosmological Λ as Zone-2 + Zone-4 substrate vacuum	T2418 + Vol 4 Ch 4
11	Non-Equilibrium Thermodynamics	Substrate cascade evolution between Zones	T2417 + Vol 4 Ch 9 cosmological cycle

12	Information Theory + Thermodynamics	Shannon-Landauer cross-link; substrate as information substrate	Vol 7 Information Theory
----	-------------------------------------	---	--------------------------

~30% existing BST coverage (Casey Wednesday principles + Vol 4 Ch 4 Λ + heat-kernel speaking pairs + T2418 vacuum unification); 70% to build at thermodynamics-first restructured framing.

Cross-volume reference map

Inputs from Vol 0: - D_IV⁵ + Casey-named principles (especially Casey “entropy = force = counting” Wednesday insight)

Inputs from Vol 1: - Casimir algebra + operator zoo + substrate-tick UV-completeness - Heat-kernel speaking pairs (Toy 273 onwards through Toy 639 k=16)

Inputs from Vol 4: - T1485 Λ + heat-kernel evaluation cross-link (Ch 5 LOAD-BEARING) - T2418 vacuum unification + T2106 eigentone gravity

Outputs to: - Vol 5 QM Ch 7 measurement formalism - Vol 7 Information Theory (entropy + Shannon) - Vol 8 Biology (thermodynamic constraints on biological substrate) - Vol 9 Condensed Matter (BCS + superconductivity)

— Vol 6 INDEX v0.1, Saturday 2026-05-23 Wave 2 scaffold